

- 8 A group of horses with the same specific characteristics is called a breed.

The photograph shows a horse of the rare breed Akhal-Teke that originated in Turkmenistan.

The Akhal-Teke population is in danger from inbreeding.



(Source: © Heat Design / Alamy Stock Photo)

- (a) A scientist suggested that the Akhal-Teke breed is the same as an older breed known as the Turkoman horse.

Describe **one** piece of information that would need to be collected in order to determine if the scientist was correct.

Give a reason for your answer.

(2)

Information

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Reason

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- (b) A scientist calculated the heterozygosity index of a population of 5 457 Akhal-Teke horses to be 0.661.

How many of these horses are heterozygotes?

(1)

- ☐ **A** 1 850
- ☐ **B** 3 607
- ☐ **C** 5 457
- ☐ **D** 8 255

- (c) The images show two other rare breeds of horse, the Caspian and the Arabian.



Caspian

(Source: © Irina Mavritsina / Alamy Stock Photo)



Arabian

(Source: © Juniors Bildarchiv GmbH / Alamy Stock Photo)

The Caspian breed was thought to be extinct until a small population was discovered in mountainous regions in 1965.

The Arabian breed is found in sandy deserts.

- (i) The Caspian breed has small hooves in proportion to its mass, whereas the closely-related Arabian breed has large hooves in proportion to its mass.

Suggest **one** reason for this difference.

(1)

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- (ii) The table shows some information about one gene for a population of 400 Caspian horses.

Genotype	two copies of the dominant allele	heterozygous	two copies of the recessive allele
Part of Hardy-Weinberg equation	p^2	$2pq$	q^2
Number of individuals	46		

Complete the table to show how many horses would have the heterozygous genotype and how many would have the homozygous recessive genotype.

Use the equation:

$$p^2 + 2pq + q^2 = 1$$

(3)



- *(d) Breeding programmes have been set up for both the Akhal-Teke and Caspian breeds.

Sperm samples have been taken from male horses from both the Akhal-Teke and Caspian populations.

Some of these male horses have since died.

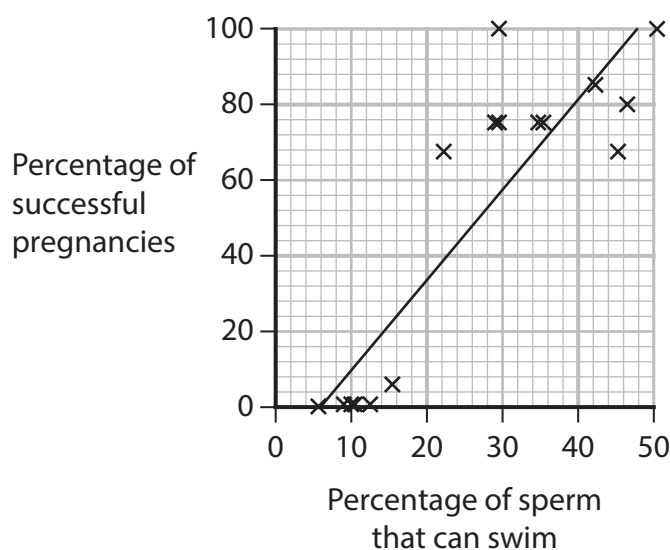
All of the sperm samples have been frozen.

The effect of freezing and thawing sperm cells was investigated.

The table shows the characteristics of two types of sperm cell samples.

Type of sample	Percentage of sperm cells that can swim	Sperm cell survival / hours	Percentage of sperm cells with damaged DNA
Fresh sperm cells	50.7	117.7	28.7
Frozen and thawed sperm cells	26.9	48.5	39.3

The graph shows the effect of the percentage of sperm cells that can swim on the number of successful pregnancies.



The correlation coefficient of the data shown in the graph is +0.90.



Discuss the advantages and disadvantages of using frozen sperm in breeding programmes for these endangered breeds.

Use all of the information in Question 8 to support your answer.

(6)

DO NOT WRITE IN THIS AREA

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(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

